

Mutual Fund Portfolio Performance & Risk Analytics

Capstone Project - Final Delivery Report

PREPARED FOR: Bluestock Fintech

PREPARED BY: Data Analyst Team

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Executive Summary

This capstone project presents an end-to-end investment analytics platform designed to evaluate mutual fund performance and monitor portfolio risk. By building an automated data engineering pipeline, we cleaned, integrated, and loaded disparate industry data files (NAV history, transaction logs, stock holdings, and fund attributes) into a centralized relational SQLite database.

Using this database, we implemented a composite weighted scorecard model that ranks funds based on risk-adjusted returns (Sharpe ratio), historical consistency (3y CAGR), manager skill (benchmark alpha), expense loads, and maximum drawdowns. Additionally, we executed advanced tail-risk calculations (VaR/CVaR), investor cohort behavioral analytics, and portfolio concentration assessments. Finally, an interactive recommender engine was deployed to dynamically match user risk appetites with optimal investment portfolios.

1. Relational Database & ETL Pipeline

To support high-performance analytical queries, a relational star schema database was implemented in SQLite. The schema isolates dimension data (dim_fund) from transactional and time-series fact data (fact_nav, fact_transactions, fact_portfolio_holdings, fact_performance, and fact_benchmark). This structure minimizes data redundancy and optimizes join speeds.

Table Name	Type	Description
dim_fund	Dimension	Fund master sheet containing fees, plans, launch dates, and fund managers.
fact_nav	Fact	Daily Net Asset Value (NAV) time series with pre-calculated daily returns.
fact_transactions	Fact	Ledger of investor transaction details including demographic information.
fact_performance	Fact	Performance analytics including Sharpe, Sortino, CAGR, and Drawdowns.
fact_portfolio_holdings	Fact	Underlying stock allocations and weights per equity fund.
fact_benchmark	Fact	Historical closing price time series of benchmark indices.

2. Data Cleaning & Quality Control

- A critical phase of the data pipeline involved cleaning and sanitizing raw source files before database insertion:
- **NAV Time-Series FFill:** Missing daily NAV records were imputed using forward-fill (FFill) grouped by AMFI code, ensuring that weekends or non-trading days did not create gaps in rolling metrics.
 - **Transaction Ledger Validation:** Audited and removed rows containing invalid non-positive transaction amounts (amount <= 0). Adjusted text representations to standard categorical representations (SIP, Lumpsum, Redemption).
 - **Plan and Option Classification:** Standardized plans (Regular vs. Direct) and options (Growth vs. Dividend) using regular expressions to resolve AMC-specific nomenclature naming errors.

3. Performance & Risk-Adjusted Return Metrics

Using the clean historical NAV series, we calculated standard performance metrics for all funds:

- **CAGR (1y/3y/5y):** Compound annual growth rates calculated peak-to-peak using actual trading days between target start and end dates.
- **Sharpe and Sortino Ratios:** Computed daily risk-adjusted ratios and annualized them using a 252-day multiplier. The risk-free rate was set at 6.5% annually. Sortino downside deviation was calculated using negative daily returns only.
- **Alpha & Beta (OLS):** Executed OLS linear regression of daily fund returns against the Nifty 100 benchmark (index_name = 'NIFTY100') to obtain beta (regression slope) and annualized alpha (intercept scaled by 252).

Top Performing Funds (Overall Weighted Scorecard)

Rank	Fund Name	Category	Sharpe	3y CAGR	Score
1	Mirae Asset Large Cap Fund	Equity	1.64	34.0%	84.5
2	ICICI Pru Midcap Fund	Equity	1.30	31.8%	83.1
3	Kotak Flexicap Fund	Equity	1.46	29.6%	81.5
4	HDFC Mid-Cap Opportunities Fund	Equity	1.18	32.4%	81.1
5	ICICI Pru Bluechip Fund	Equity	1.11	32.5%	80.5

4. Advanced Risk & Behavioral Analytics

To deepen our risk assessment, advanced metrics were computed across the database:

4.1 Value at Risk (VaR) & Conditional VaR (CVaR)

Value at Risk (95% daily VaR) identifies the threshold loss that is exceeded only 5% of the time. Conditional VaR (CVaR) represents the expected average return on those worst 5% of trading days. Equity funds exhibited substantial downside tails (VaR ranging from -1.8% to -2.3% daily), whereas debt and liquid funds maintained extremely tight profiles (VaR < -0.15% daily), validating their role as wealth-preservation vehicles.

4.2 Sector Concentration (HHI) for Equity Funds

The Herfindahl-Hirschman Index (HHI) was calculated across all equity funds based on their underlying sector weights. Equity portfolios with an HHI below 1500 are highly diversified, while funds exceeding 2500 exhibit significant concentration (sector-focused portfolios). This allows advisors to flag concentration risk in thematic portfolios.

4.3 Investor Behavioral Cohort Analysis

Cohort Year	Investors	Avg SIP (INR)	Total Invested (Cr)	Preferred Fund
2024	4,803	Rs. 10,996.89	Rs. 225.81 Cr	Mirae Asset Emerging Bluechip Fund
2025	197	Rs. 13,505.21	Rs. 1.90 Cr	SBI Small Cap Fund

4.4 SIP Continuity & At-Risk Accounts

For clients with 6 or more monthly SIP transactions, we tracked the average interval (gap) between consecutive payment dates. If an investor's average gap exceeds 35 days, they are flagged as an 'at-risk' account showing signs of payment default. Out of 1,362 eligible SIP clients, 1,332 (97.8%) were flagged as at-risk. This high rate underlines a critical operational vulnerability where payment mandates are either failing or not being consistently renewed.

5. Strategic Advisory & Operational Recommendations

- **Address SIP Mandate Defaults:** Operational teams should immediately establish automated email and SMS alerts 5 days prior to SIP dates to reduce the 97.8% default risk.
- **Push Direct Plans:** Financial advisors must highlight the compounding benefit of Direct Plans. Direct plans skip broker commissions, generating significantly higher Sharpe ratios and long-term AUM.
- **Use Scorecard Benchmarking:** Fund research teams should replace simple return-ranking systems with the composite scorecard, capturing tail risk (Drawdown) and cost (Expense) alongside absolute performance.